

Systems-Level Reliability and Robustness Evaluation Framework for Document AI

Master's Thesis Defense

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Introduction / Problem Statement

Document AI Complexity: Modern enterprise pipelines rely heavily on automated extraction of information from complex PDF documents.

The PDF Challenge: Unstructured documents contain rich multi-column formats, dense tabular grids, and geometric layouts that defy simple reading orders.

Hallucination & Layout Risks: Semantic understanding without visual awareness leads to layout-blindness, causing critical extraction failures.

Critical Downstream Risk: A tiny character reading error (e.g., distorting **\$1,240.50** into a rounded **\$1,200.00** on a scanned invoice) is not just a simple spelling mistake—it propagates into automated financial ledgers, causing catastrophic audit and compliance failures in finance and healthcare.

Research Objective

Addressing the Perception-Cognition Gap: Bridging the disconnect where LLMs show advanced linguistic reasoning but fail at structural, spatial perception.

Scientific Goal: Formulating a rigorous, systems-level reliability evaluation framework for DocVQA architectures under zero-shot conditions.

Robustness Benchmarking: Swapping and analyzing perception modules to establish a reliable accuracy-efficiency frontier for real-world deployments.

Traditional OCR (Tesseract)

Heuristic Baseline: Relies on classical layout algorithms to group and serialize characters.

Linearization Fragmentation: Converts multi-column text horizontally into a single continuous stream, destroying reading order and paragraph alignment.

Fidelity Loss: Separates tabular cells from headers, making spatial relation reasoning impossible.

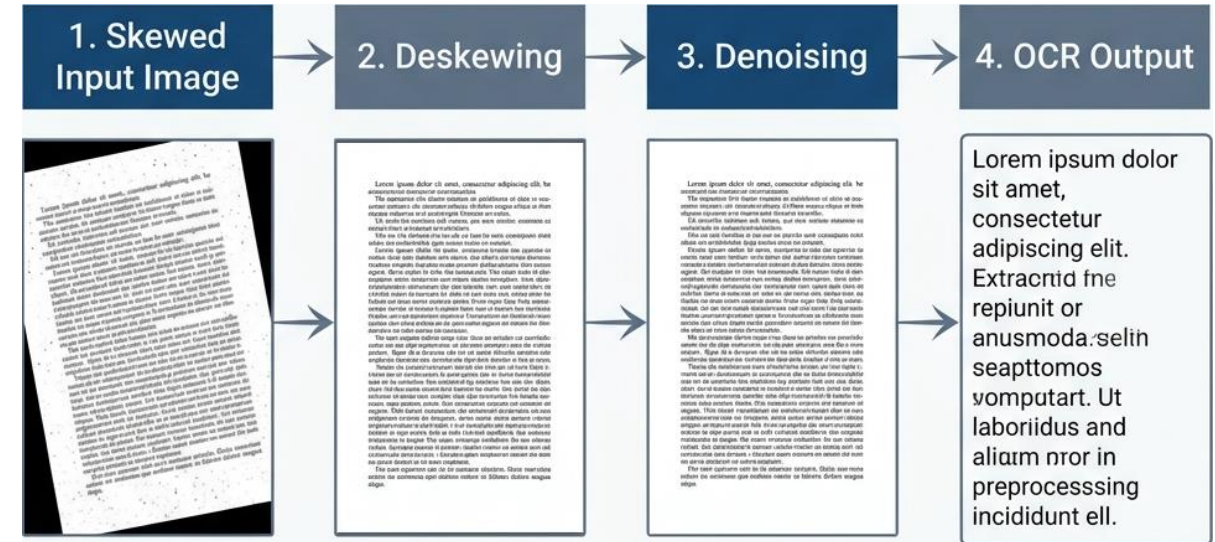


Figure 4.3: OCR Preprocessing Pipeline

PaddleOCR

Deep-Learning Spatial Detection: Employs an academic DBNet detector combined with SVTR recognizer for raw character mapping.

Layout Robustness: Highly precise alphanumeric localization, minimizing character confusion.

The Cognitive Limit: Captures characters perfectly but lacks semantic context, requiring a downstream layout reconstruction layer.

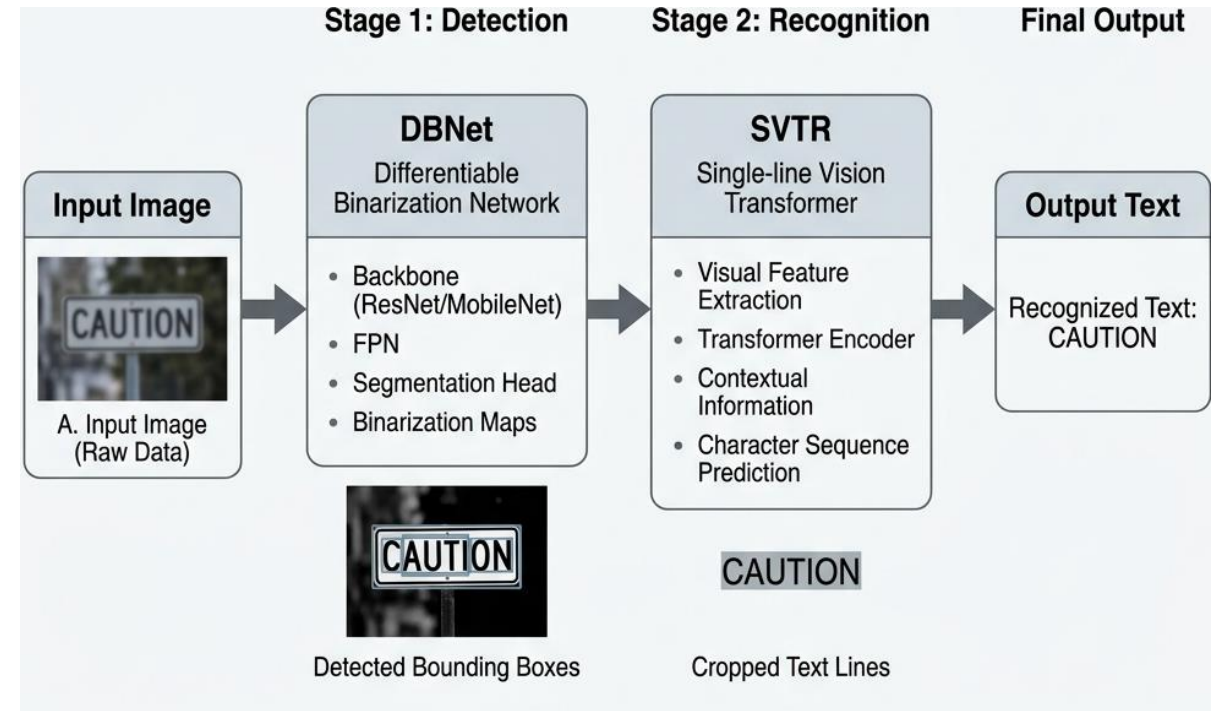


Figure 2.2: PaddleOCR Advanced Multi-Stage Architecture

Perception Layer

Layout Detection & Text Localization: Automatically mapping page regions, paragraphs, and nested table cells.

Character Grounding: Indexing absolute 2D bounding boxes (x_1, y_1, x_2, y_2) to ensure alphanumeric fidelity.

Spatial Alignment: Preparing structured, geometry-preserved raw text streams for cognitive processing.

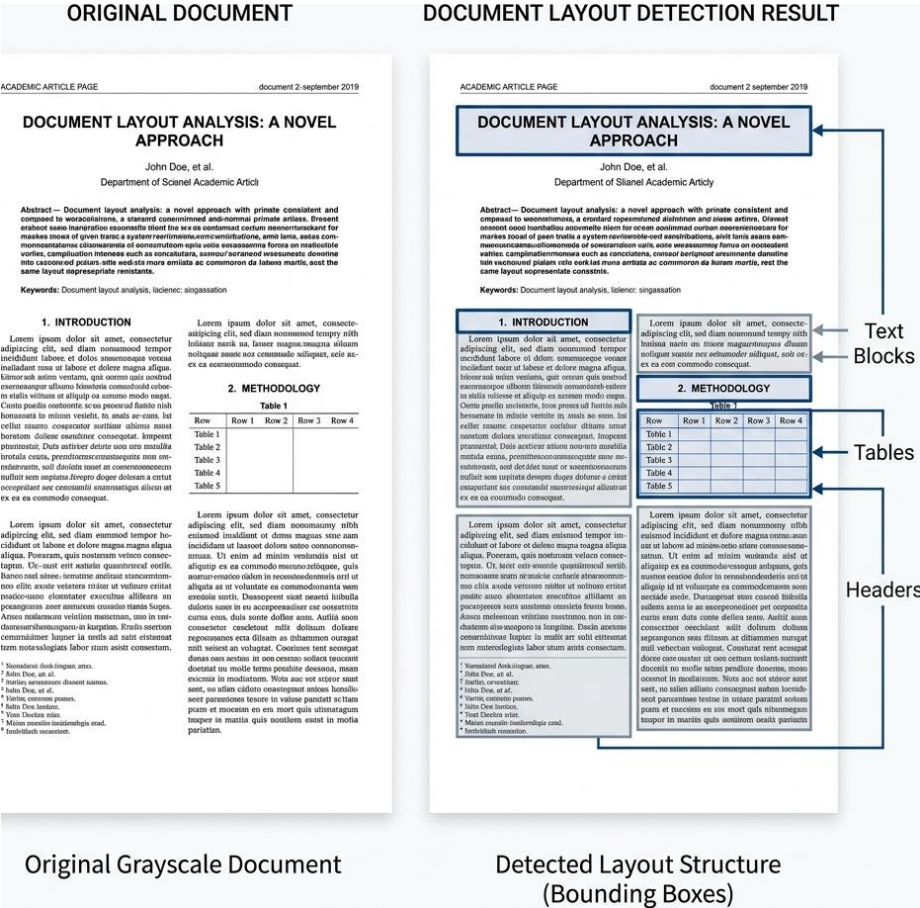


Figure 2.3: Layout Detection and Document Structure Logic

Vision-Language Models (VLMs)

Multimodal Generation: Treats document images natively as input patches, removing the need for heuristic pipelines.

Resolution-Loss Hallucination: Fixed patch resolutions (e.g., 336x336 pixels) compress fine-grained tabular data, leading to character distortion.

The Perception Failure: "Probabilistic guessing" of dense digits in complex zero-shot layouts due to resolution-loss and pixel alignment errors.

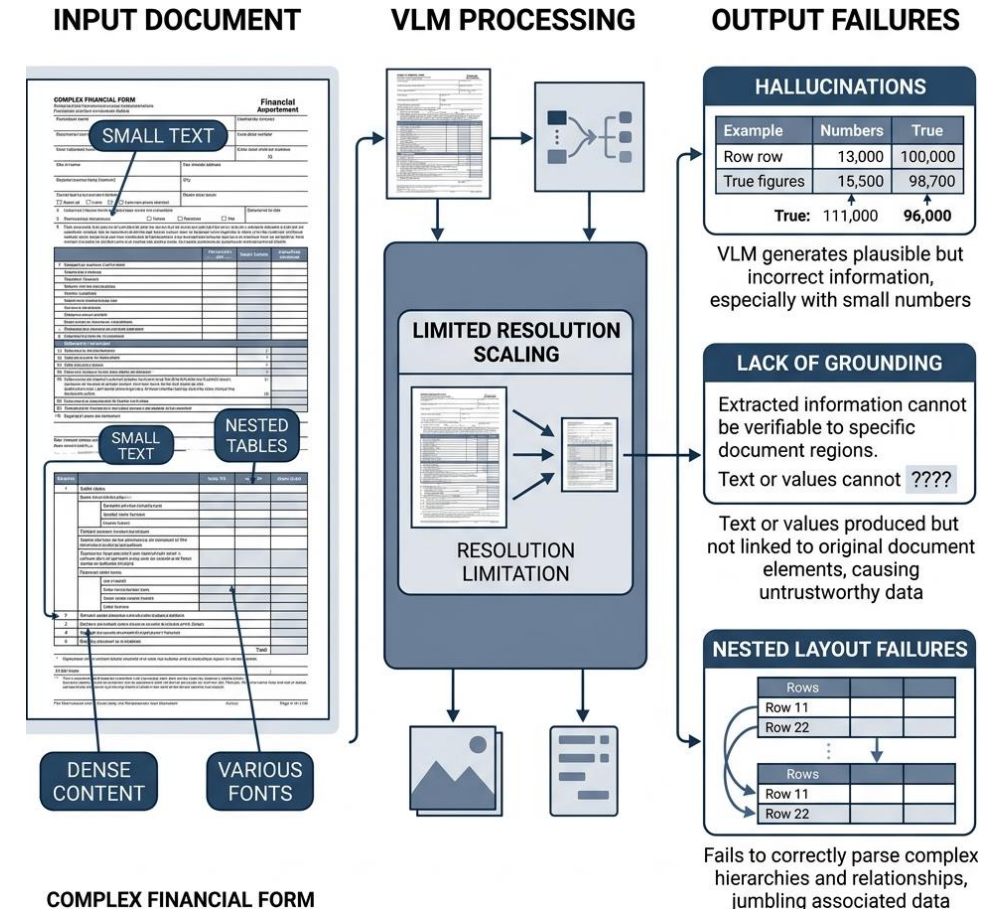


Figure 2.4: VLM Projection Layer and Resolution Constraints

Hybrid OCR-VLM Synchronization

Dual-Stream Synchronization: Merging the literal precision of PaddleOCR with the high-level semantic layout awareness of VLMs.

Perception Safety Net: Grounding VLM spatial reasonings inside verified deterministic OCR character coordinates.

Robustness Fusing: Drastically reduces layout confusion and generative hallucinations in zero-shot extractions.

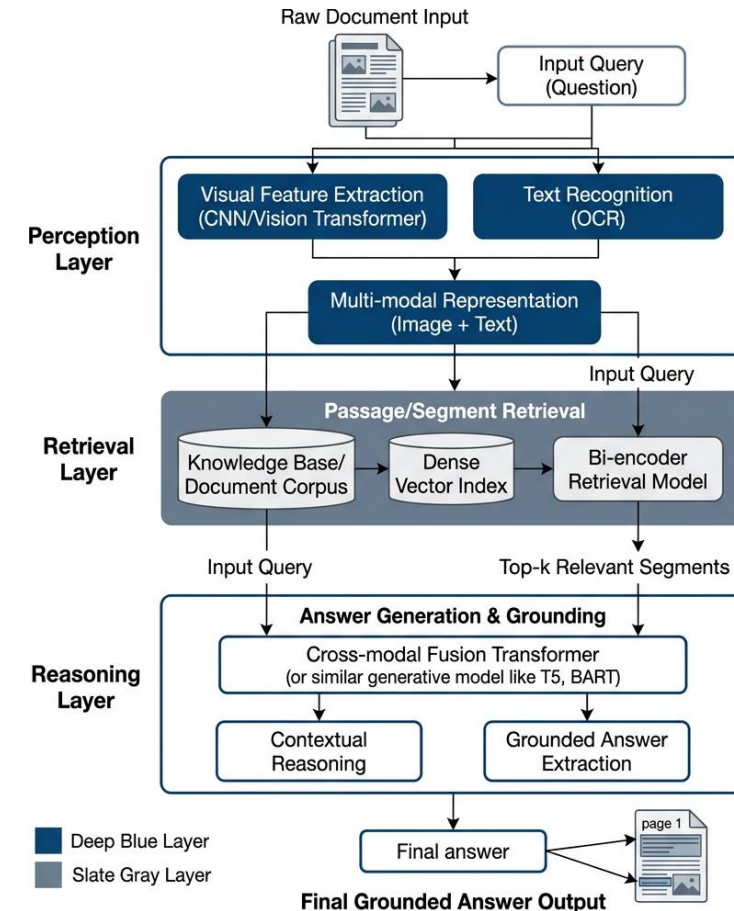


Figure 4.1: Advanced Global System Orchestration Architecture

RAG / Connection Layer

Vector Space Mapping: Projecting layout-aware chunked documents into a FAISS vector database using `all-MiniLM-L6-v2`.

Hybrid Context Retrieval: Querying and fetching high-relevance chunks grounded with precise spatial metadata.

Cognitive Reasoning: Supplying localized context to the Mistral 7B Instruct reasoning engine.

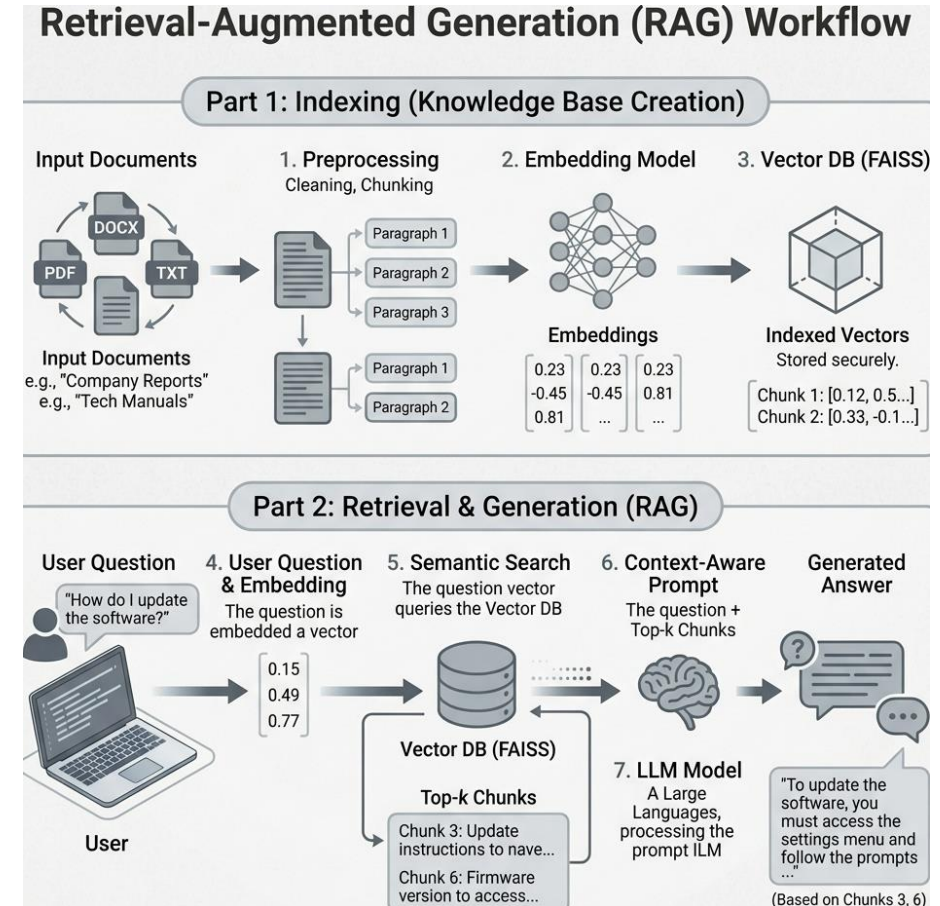


Figure 3.1: Semantic Embedding and Vector Storage Workflow

Benchmark Framework

Test Corpus: Compiled 50 high-complexity PDFs from the DocVQA validation set (dense nested tables, scan noise, multi-column designs).

Strict Zero-Shot Paradigm: No domain-specific or layout-specific fine-tuning, evaluating raw generalization robustness.

Swappable Architecture: Systematically isolating and measuring the impact of each perception stream under identical reasoning conditions.

Evaluation Metrics

N: Total size of evaluation corpus (50 document queries)

$$ANLS = \frac{1}{N} \sum_{i=1}^N s(a_i, g_i) \quad (3.1)$$

a_i: Predicted answer text generated by the cognitive model

$$EM = \mathbf{1}(a_i = g_i) \quad (3.2)$$

g_i: Authoritative human-verified ground truth answer

$$F1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (3.3)$$

s(a_i, g_i): Similarity score with 0.5 threshold (similarity < 0.5 is scored 0)

EM: Exact Match (absolute binary character-level identity)

$$T_p = \frac{N}{\sum L_i} \quad (3.5)$$

F1: F1-Score (harmonic mean of token-level Precision P and Recall R)

Experimental Results

Empirical Breakthrough: The proposed Hybrid model achieves the highest accuracy, outperforming standalone VLMs and OCR pipelines.

The Performance Frontier: Validating that visual layout synchronization is essential to resolve complex DocVQA prompts.

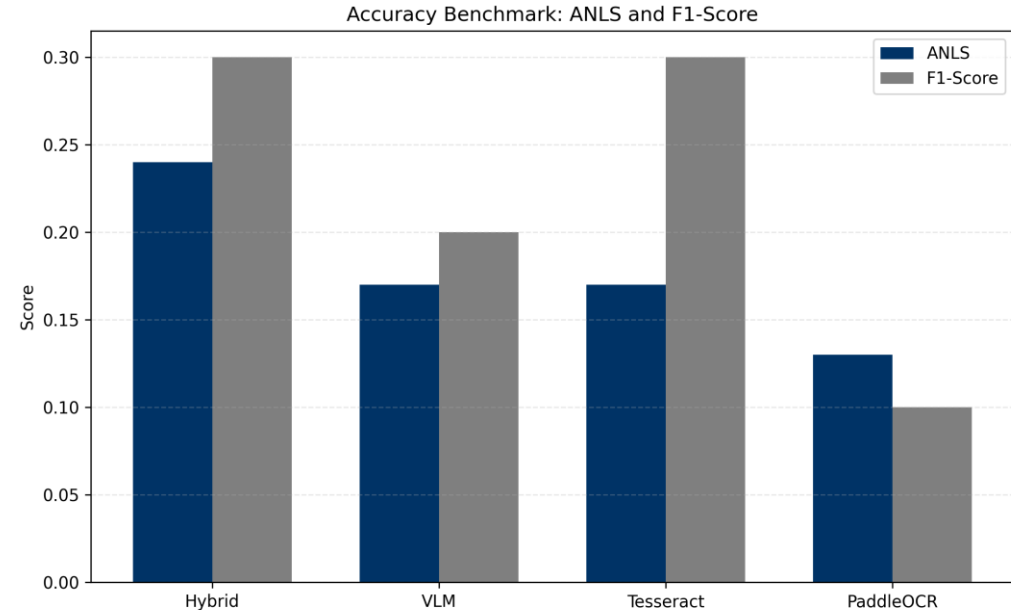


Figure 7.2: Accuracy Benchmark Matrix (ANLS vs F1)

Performance Metrics Table

Comprehensive Evaluation: Performance comparison across accuracy, operational latency, throughput, and system resource overhead.

Accuracy vs. Efficiency: The Hybrid model achieves the highest accuracy at the cost of higher latency and RAM consumption.

System Model	ANLS	Exact Match (EM)	F1-Score	Inference Latency	Throughput	RAM Footprint	Search Efficiency
Hybrid (Ours)	0.24	0.20	0.30	14.20 s	0.07 S/s	4600 MB	Sub-ms (<1ms)
Standalone VLM	0.17	0.10	0.20	4.20 s	0.24 S/s	4100 MB	N/A
Tesseract RAG	0.17	0.10	0.30	11.00 s	0.09 S/s	350 MB	Sub-ms (<1ms)
PaddleOCR RAG	0.13	0.00	0.10	52.30 s	0.02 S/s	850 MB	Sub-ms (<1ms)

Table 7.1: Exhaustive Performance Benchmarking Matrix

Ablation Study

Isolating Stream Contributions: Systematic removal of layout spatial detection and character grounding streams.

Perception Grounding Efficacy: Proves that character coordinate anchoring is the primary driver in reducing hallucination rates.

Architecture	Layout Detection	Text Grounding	VLM Context	ANLS	EM
Hybrid (Full)	Yes	Yes	Yes	0.24	0.20
No Grounding	Yes	No	Yes	0.17	0.10
No Layout	No	Yes	Yes	0.13	0.00

Table 7.2: Perception Component Ablation Study

Error Analysis / Case Studies

Low ANLS Failure Modes: Poor document contrast, low scanning resolution, and heavily overlapping text boundaries.

Visual Evidence: VLM resolution-loss vs. Hybrid character-grounding precision.

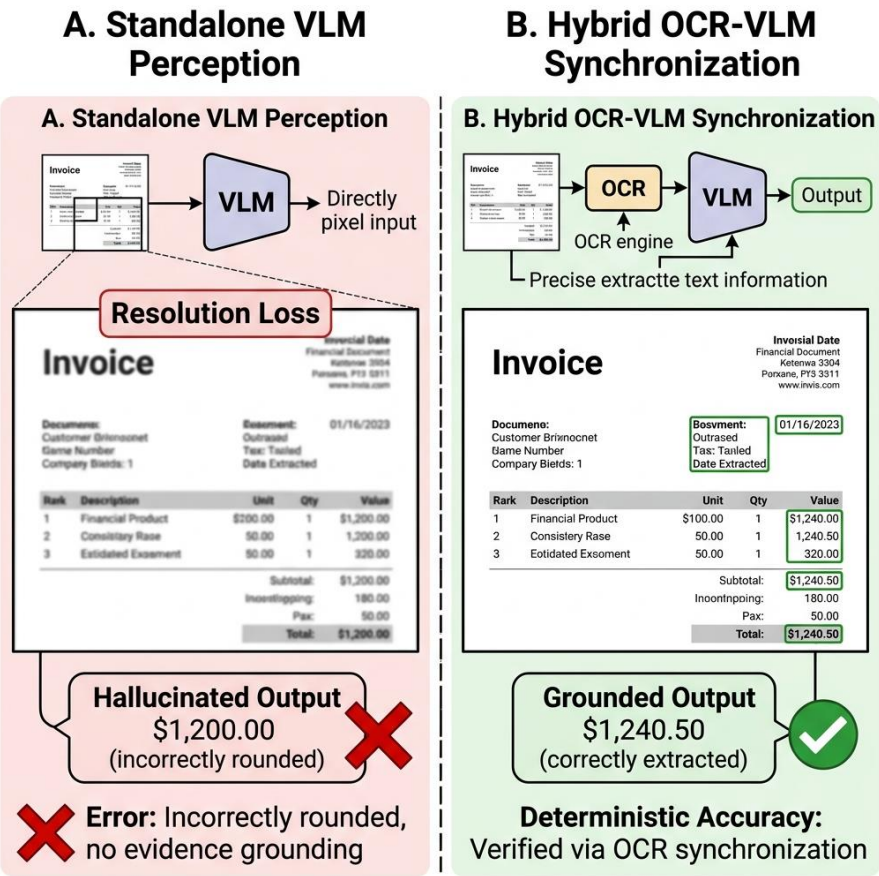


Figure 7.6: Visual Case Study of VLM Resolution-Loss vs. Hybrid Precision

Deployment Considerations

Accuracy-Efficiency Trade-off: High-reliability synchronization demands dual neural network overhead (14.2s latency).

Memory Footprint: Fusing multiple streams increases RSS RAM footprint (4600MB).

Recommendation: Deploy in offline mission-critical pipelines, using async processing to mitigate CPU-bound latency.

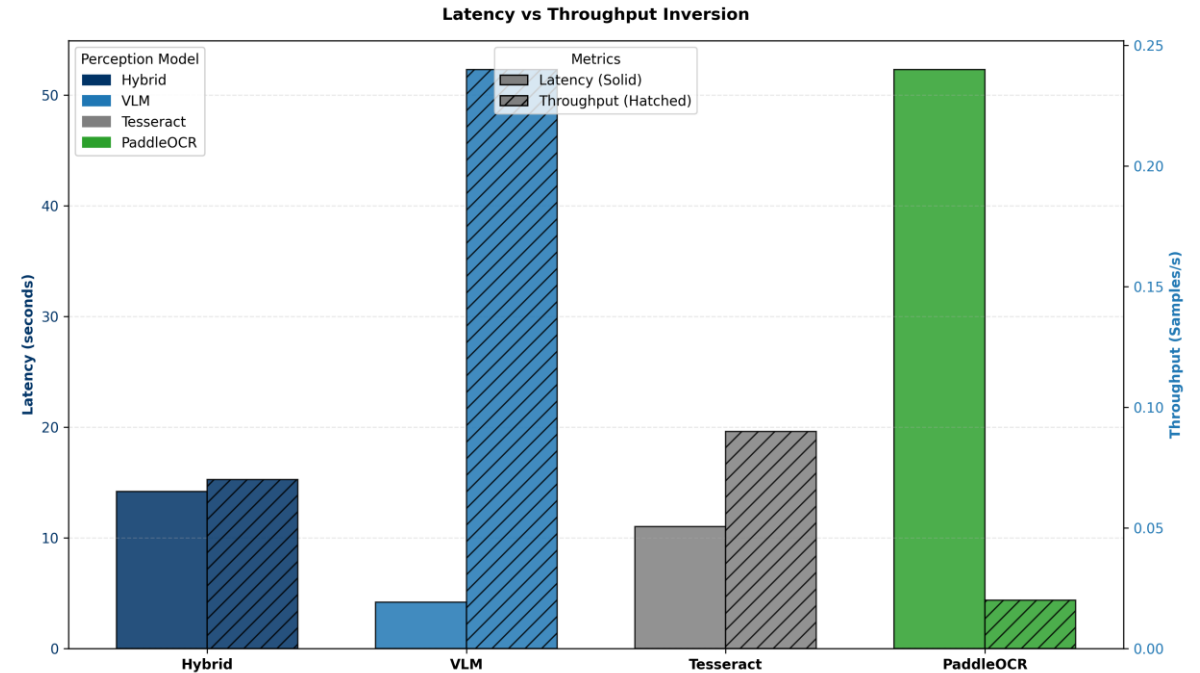


Figure 7.3: Accuracy-Efficiency Frontier and Latency Inversion

Conclusion

Bridging the Gap: Synchronized OCR-VLM architecture successfully bridges the perception-cognition disconnect.

Grounding Efficacy: Deterministic character grounding is vital for eliminating LLM hallucinations in structured documents.

Standardized Framework: Provides a repeatable system-level benchmark for modern Document AI research.

Future Work: Multi-Agent System

Agentic Pipeline Evolution: Transitioning from a rigid pipeline to a cooperative multi-agent network.

Layout Detection Agent: Responsible for spatial column splits, tabular boundary parsing, and geometric structure.

Text Detection & OCR Agent: Specialized in high-fidelity character recognition, raw transcription, and bounding box coordinates.

Image & Visual Perception Agent: Dedicated to understanding non-textual primitives, charts, figures, and embedded artwork.

Coordinator & Routing Agent: Orchestrating agent dialogue, resolving data conflicts, and synthesizing unified factual contexts.

Thank You

Thank You for Your Attention
Questions & Discussion
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